

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1\alpha\beta\chi}$				
$\mathcal{K}_{0^+}^{\#1} + \frac{1}{2}(-\gamma_1 k^2 - 3 \binom{(2)}{k_3})$	0					
$\mathcal{K}_{0^+}^{\#1} + \alpha\beta\chi$	0	$\gamma_2 k^2$	$\mathcal{K}_{1^+}^{\#1\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1\alpha}$	$\mathcal{K}_{1^+}^{\#2\alpha}$
$\mathcal{K}_{1^+}^{\#1} + \alpha\beta$	0	0	0	0		
$\mathcal{K}_{1^+}^{\#2} + \alpha\beta$	0	0	0	0		
$\mathcal{K}_{1^+}^{\#1} + \alpha$	0	0	$\frac{1}{2}(\gamma_3 k^2 - 2 \binom{(2)}{k_3})$	$\frac{\gamma_4 k^2 + 2 \binom{(2)}{k_3}}{2 \sqrt{2}}$		
$\mathcal{K}_{1^+}^{\#2} + \alpha$	0	0	$\frac{\gamma_4 k^2 + 2 \binom{(2)}{k_3}}{2 \sqrt{2}}$	$\frac{1}{2}(-\gamma_5 k^2 - \binom{(2)}{k_3})$	$\mathcal{K}_{2^+}^{\#1\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1\alpha\beta\chi}$
			$\mathcal{K}_{2^+}^{\#1} + \alpha\beta$	$\gamma_6 k^2$	0	
			$\mathcal{K}_{2^+}^{\#1} + \alpha\beta\chi$	0	$\frac{\gamma_7 k^2}{2}$	

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^-}^{\#1} a\beta_X$				
$\mathcal{T}_{0^+}^{\#1} \dagger$	$\frac{1}{\text{Det}(0^+)}$	0				
$\mathcal{T}_{0^+}^{\#1} \dagger^{a\beta_X}$	0	$\frac{1}{k^2 \binom{(4)}{\kappa_{15} \kappa_{16}}}$	$\mathcal{T}_{1^+}^{\#1} a\beta$	$\mathcal{T}_{1^+}^{\#2} a\beta$	$\mathcal{T}_{1^+}^{\#1} \alpha$	$\mathcal{T}_{1^+}^{\#2} \alpha$
	$\mathcal{T}_{1^+}^{\#1} \dagger^{a\beta}$	0	0	0	0	0
	$\mathcal{T}_{1^+}^{\#2} \dagger^{a\beta}$	0	0	0	0	0
	$\mathcal{T}_{1^+}^{\#1} \dagger^\alpha$	0	0	$\frac{-Y_5 k^2 - 2 \binom{(2)}{\kappa_3}}{2 \text{Det}(1^+)}$	$\frac{-Y_4 k^2 - 2 \binom{(2)}{\kappa_3}}{2 \sqrt{2} \text{Det}(1^+)}$	
	$\mathcal{T}_{1^+}^{\#2} \dagger^\alpha$	0	0	$\frac{-Y_4 k^2 - 2 \binom{(2)}{\kappa_3}}{2 \sqrt{2} \text{Det}(1^+)}$	$\frac{Y_3 k^2 - 2 \binom{(2)}{\kappa_3}}{2 \text{Det}(1^+)}$	$\mathcal{T}_{2^+}^{\#1} a\beta$ $\mathcal{T}_{2^+}^{\#1} a\beta_X$
			$\mathcal{T}_{2^+}^{\#1} \dagger^{a\beta}$	$\frac{1}{\text{Det}(2^+)}$	0	
			$\mathcal{T}_{2^+}^{\#1} \dagger^{a\beta_X}$	0	$\frac{1}{\text{Det}(2^+)}$	

$$\begin{aligned}
 & \text{Abbreviations used in matrices} \\
 & Y_1 = 3 \binom{(4)}{\kappa_1} - 4 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_2} + 2 \binom{(4)}{\kappa_3} \& Y_2 = \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} \& Y_3 = 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_2} \& \\
 & Y_4 = 2 \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_7} \& Y_5 = \binom{(4)}{\kappa_{16}} + \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_7} \& Y_6 = 2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_3} \& \\
 & Y_7 = 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \& \text{Det}(0^+) = \frac{1}{2} (-k^2 (3 \binom{(4)}{\kappa_1} - 4 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_2} + 2 \binom{(4)}{\kappa_3}) - 3 \binom{(2)}{\kappa_3}) \& \\
 & \text{Det}(1^-) = \frac{1}{8} (-k^4 (2 \binom{(4)}{\kappa_{16}}^2 - 4 \binom{(4)}{\kappa_2} \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_7}^2 + 2 \binom{(4)}{\kappa_{16}} (-\binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_7}) + 4 \binom{(4)}{\kappa_{15}} (\binom{(4)}{\kappa_{16}} + \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_7})) + \\
 & 2k^2 (-2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} + 2 \binom{(4)}{\kappa_3}) \binom{(2)}{\kappa_3}) \& \text{Det}(2^+) = k^2 (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_3})
 \end{aligned}$$

$$\begin{aligned} & \text{Lagrangian} \\ & \kappa_3^{(2)} \mathcal{K}_\alpha^\alpha \beta \mathcal{K}_{\beta\chi}^\chi + \kappa_1^{(4)} \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^\chi \mathcal{K}_\alpha^\alpha \beta + \kappa_2^{(4)} \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^\chi \mathcal{K}_\alpha^\alpha \beta + \kappa_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \\ & 2 \kappa_{16}^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 2 \kappa_{15}^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + \kappa_{16}^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + \kappa_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + \\ & \kappa_7^{(4)} \partial^\chi \mathcal{K}_\alpha^\alpha \beta \partial_\delta \mathcal{K}_{\chi\beta}^\delta - \kappa_{15}^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \kappa_{15}^{(4)} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \kappa_{16}^{(4)} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{a\beta\chi} \mathcal{J}_{a\beta\chi}$
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[illegible]

$\mathcal{I}_{1^+}^{\#2\alpha\beta} = 0$	3	$\partial_\delta \partial_\chi \partial^\alpha \mathcal{I}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{I}^{\chi\alpha\beta} = \partial_\delta \partial_\chi \partial^\beta \mathcal{I}^{\chi\alpha\delta}$
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$\mathcal{F}_1^{\#1ab} = 0$	3	$\partial_\delta \partial_\chi \partial^a \mathcal{F}^{\chi p \delta} + \partial_\delta \partial^{\delta \delta} \partial_\chi \mathcal{F}^{p a \chi} = \partial_\delta \partial_\chi \partial^{\delta \delta} \mathcal{F}^{\chi a \delta} + \partial_\delta \partial^{\delta \delta} \partial_\chi \mathcal{F}^{a p \chi}$
$\mathcal{F}_1^{\#1ab} = 0$	5	

Total # constraint(s):	6		
Resolved pole(s)	# polarization(s)	Square mass	Residue

Resolved pole(s)	n polarization(s)	Square mass	Residue
			$\begin{matrix} (4) & (4) & (4) \end{matrix}$

	1	0	$-\frac{6\kappa_{15}+\kappa_{16}-2\kappa_3}{(2\kappa_{15}+\kappa_{16})(2\kappa_{15}-\kappa_3)}$
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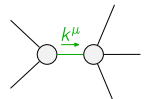
	1	0	$\frac{6 \kappa_{15}^{(4)} + \kappa_{16}^{(4)} - 2 \kappa_3^{(4)}}{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (2 \kappa_{15}^{(4)} - \kappa_3^{(4)})}$
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	2	0	$-\frac{16 \kappa_{15}^{(4)2} + \kappa_{16}^{(4)2} + 4 \kappa_{15}^{(4)} (\kappa_{16}^{(4)} \kappa_3^{(4)}) + 3 \kappa_3^{(4)2}}{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (2 \kappa_{15}^{(4)} \kappa_{16}^{(4)} - 2 \kappa_3^{(4)}) (2 \kappa_{15}^{(4)} \kappa_3^{(4)})}$
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$$-((-16 \binom{(4)}{K_{15}} \binom{(2)}{K_3} + 2 \binom{(4)}{K_{16}} \binom{(2)}{K_3} \binom{(4)}{K_3} + 8 \binom{(4)}{K_{16}} \binom{(4)}{K_3} \binom{(2)}{K_3} +$$

	1	0	$\frac{8 \kappa_3^2 \kappa_3 + 8 \kappa_{15}^2 (2 \kappa_{16} + 5 \kappa_3) \kappa_3 - 4 \kappa_{15}^2 (\kappa_{16} + 6 \kappa_{16} \kappa_3 + 8 \kappa_3) \kappa_3 + \sqrt{(-2 \kappa_{15}^2 + \kappa_{16}^2 + 2 \kappa_{15}^2 (4 \kappa_{15}^2 + \kappa_{16}^2 + 4 \kappa_{15}^2 (\kappa_{16} - 12 \kappa_3) + 12 \kappa_3^2 (2 \kappa_{15}^2) / ((2 \kappa_{15}^2 + \kappa_{16}^2) (2 \kappa_{15}^2 - \kappa_3) (-2 \kappa_{15}^2 + \kappa_{16}^2 + 2 \kappa_3^2)^2 \kappa_3))}}{2 \kappa_{15}^2 \kappa_3^2}$
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	1	0	$\frac{(16 \binom{4}{15} \binom{2}{3} - 2 \binom{4}{16} \binom{2}{3} - 8 \binom{4}{16} \binom{2}{3} - 8 \binom{4}{3} \binom{2}{3} - 8 \binom{4}{15} \binom{2}{3} + 5 \binom{4}{3} \binom{2}{3} + 4 \binom{4}{15} \binom{2}{3} + 6 \binom{4}{16} \binom{2}{3} + 8 \binom{4}{3} \binom{2}{3})}{\sqrt{(-2 \binom{4}{15} + \binom{4}{16} + 2 \binom{4}{3})^4 (52 \binom{4}{15}^2 + \binom{4}{16}^2 + 4 \binom{4}{15} \binom{4}{16} - 12 \binom{4}{3}) + 12 \binom{4}{3}^2 \binom{4}{15}^2}} / ((2 \binom{4}{15} + \binom{4}{16}) (2 \binom{4}{15} - \binom{4}{3}) (-2 \binom{4}{15} + \binom{4}{16} + 2 \binom{4}{3})^2 \binom{2}{3})$
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	1	0	$ \begin{aligned} & -((-128 \kappa_{15}^{(4)} p^2 + 32 \kappa_{16}^{(4)} \kappa_3^{(4)} p^2 + 32 \kappa_{16}^{(4)} \kappa_3^{(4)} p^2 + \\ & 8 \kappa_{16}^{(4)} \kappa_3^{(4)} p^2 + 48 \kappa_{16}^{(4)} \kappa_3^{(4)} \kappa_7 p^2 + 24 \kappa_{16}^{(4)} \kappa_3^{(4)} \kappa_7 p^2 + 18 \kappa_{16}^{(4)} \kappa_3^{(4)} \kappa_7 p^2 - \\ & 64 \kappa_{15}^{(4)} (5 \kappa_{16}^{(4)} + \kappa_3^{(4)} + 3 \kappa_7^{(4)}) p^2 - 8 \kappa_{15}^{(4)} (32 \kappa_{16}^{(4)} - 4 \kappa_3^{(4)} + 9 \kappa_7^{(4)} + 4 \kappa_{16}^{(4)} (\kappa_3^{(4)} + 9 \kappa_7^{(4)})) p^2 - \\ & 4 \kappa_{15}^{(4)} (16 \kappa_{16}^{(4)} - 8 \kappa_{16}^{(4)} (2 \kappa_3^{(4)} - 3 \kappa_7^{(4)}) - \kappa_3^{(4)} (2 \kappa_3^{(4)} + 3 \kappa_7^{(4)})^2 + \kappa_{16}^{(4)} (-20 \kappa_3^{(4)} - 24 \kappa_3^{(4)} \kappa_7 + 9 \kappa_7^{(4)})) p^2 + \\ & \sqrt{3} \sqrt{((2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)})^2 (4096 \kappa_{15}^{(4)6} p^4 + 4096 \kappa_{15}^{(4)5} (2 \kappa_{16}^{(4)} - \kappa_3^{(4)} + 2 \kappa_7^{(4)}) p^4 + \kappa_{16}^{(4)4} (3 \kappa_3^{(2)2} + 512 \kappa_3^{(4)2} p^4) + \\ & 4 \kappa_{16}^{(4)3} \kappa_3^{(4)} (15 \kappa_3^{(2)2} + 64 \kappa_3^{(4)} (6 \kappa_3^{(4)} + 5 \kappa_7^{(4)}) p^4) + 16 \kappa_{15}^{(4)4} (147 \kappa_3^{(2)2} + 64 (6 \kappa_{16}^{(4)2} - 8 \kappa_{16}^{(4)} \kappa_3^{(4)} + \kappa_3^{(4)2} + \\ & 12 \kappa_{16}^{(4)} \kappa_7 - 8 \kappa_3^{(4)} \kappa_7 + 6 \kappa_7^{(4)}) p^4) + 12 \kappa_{16}^{(4)2} \kappa_3^{(4)2} (33 \kappa_3^{(4)2} + 8 (20 \kappa_3^{(4)} + 28 \kappa_3^{(4)} \kappa_7 + 13 \kappa_7^{(4)}) p^4) + \\ & 4 \kappa_3^{(4)2} (48 \kappa_3^{(4)4} p^4 + 160 \kappa_3^{(4)3} \kappa_7 p^4 + 72 \kappa_3^{(4)2} \kappa_7^2 p^4 + 27 \kappa_7^{(4)4} p^4 + 24 \kappa_3^{(4)2} (8 \kappa_3^{(2)2} + 7 \kappa_7^{(4)2} p^4)) + \\ & 64 \kappa_{16}^{(4)} \kappa_3^{(4)2} (16 \kappa_3^{(4)3} p^4 + 36 \kappa_3^{(4)2} \kappa_7 p^4 + 9 \kappa_7^{(4)3} p^4 + 3 \kappa_3^{(4)} (5 \kappa_3^{(2)2} + 8 \kappa_7^{(4)2} p^4)) + \\ & 32 \kappa_{15}^{(4)3} (-84 \kappa_{16}^{(4)} \kappa_3^{(4)2} + 128 \kappa_{16}^{(4)3} p^4 + 64 \kappa_3^{(4)3} p^4 + 288 \kappa_{16}^{(4)2} \kappa_7 p^4 + 160 \kappa_3^{(4)2} \kappa_7 p^4 + \\ & 72 \kappa_7^{(4)3} p^4 + 16 \kappa_{16}^{(4)} (16 \kappa_3^{(4)2} - 12 \kappa_3^{(4)} \kappa_7 + 15 \kappa_7^{(4)2}) p^4 - 3 \kappa_3^{(4)} (77 \kappa_3^{(4)2} + 48 \kappa_7^{(4)2} p^4)) + \\ & 8 \kappa_{15}^{(4)2} (256 \kappa_{16}^{(4)4} p^4 + 128 \kappa_{16}^{(4)3} (2 \kappa_3^{(4)} + 5 \kappa_7^{(4)}) p^4 + \kappa_{16}^{(4)} (738 \kappa_3^{(4)2} - 256 \kappa_3^{(4)2} p^4 + 768 \kappa_3^{(4)2} \kappa_7 p^4 - \\ & 192 \kappa_3^{(4)} \kappa_7 p^4 + 288 \kappa_7^{(4)3} p^4) + 3 \kappa_{16}^{(4)2} (39 \kappa_3^{(2)2} + 16 (8 \kappa_3^{(4)2} + 4 \kappa_3^{(4)} \kappa_7 + 13 \kappa_7^{(4)2}) p^4) + 2 (-80 \kappa_3^{(4)4} p^4 - \\ & 32 \kappa_3^{(4)3} \kappa_7 p^4 - 72 \kappa_3^{(4)2} \kappa_7^2 p^4 + 27 \kappa_7^{(4)4} p^4 + 3 \kappa_3^{(4)2} (177 \kappa_3^{(2)2} + 56 \kappa_7^{(4)2} p^4)) - 8 \kappa_{15}^{(4)} (256 \kappa_{16}^{(4)4} \kappa_3 p^4 + \\ & 4 \kappa_{16}^{(4)3} (3 \kappa_3^{(2)2} + 160 \kappa_3^{(4)} (\kappa_3^{(4)} + \kappa_7^{(4)}) p^4) + 3 \kappa_{16}^{(4)2} \kappa_3^{(4)2} (51 \kappa_3^{(2)2} + 16 (16 \kappa_3^{(4)2} + 22 \kappa_3^{(4)} \kappa_7 + 13 \kappa_7^{(4)2}) p^4) + \\ & 2 \kappa_3^{(4)} (16 \kappa_3^{(4)4} p^4 + 112 \kappa_3^{(4)3} \kappa_7 p^4 + 36 \kappa_3^{(4)2} \kappa_7^2 p^4 + 27 \kappa_7^{(4)4} p^4 + 24 \kappa_3^{(4)2} (11 \kappa_3^{(4)2} + 6 \kappa_7^{(4)2} p^4)) + \\ & 2 \kappa_{16}^{(4)} \kappa_3^{(4)} (160 \kappa_3^{(4)3} p^4 + 480 \kappa_3^{(4)2} \kappa_7 p^4 + 144 \kappa_7^{(4)3} p^4 + 3 \kappa_3^{(4)} (87 \kappa_3^{(2)2} + 88 \kappa_7^{(4)2} p^4)))))) / \\ & ((2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (2 \kappa_{15}^{(4)} - \kappa_3^{(4)}) (-2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)} + 2 \kappa_3^{(4)} (2 \kappa_3^{(2)2} + \kappa_3^{(4)2}))) \end{aligned} $
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	1	0	
			$\begin{aligned} & (128 \binom{(4)}{\kappa_{15}} p^2 - 32 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} p^2 - 32 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} p^2 - \\ & 8 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} p^2 - 48 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7} p^2 - 24 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7} p^2 - 18 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7} p^2 + \\ & 64 \binom{(4)}{\kappa_{15}} (5 \binom{(4)}{\kappa_{16}} + \binom{(4)}{\kappa_3} + 3 \binom{(4)}{\kappa_7}) p^2 + 8 \binom{(4)}{\kappa_{15}} (32 \binom{(4)}{\kappa_{16}} - 4 \binom{(4)}{\kappa_3} + 9 \binom{(4)}{\kappa_7} + 4 \binom{(4)}{\kappa_{16}} (\binom{(4)}{\kappa_3} + 9 \binom{(4)}{\kappa_7})) p^2 + \\ & 4 \binom{(4)}{\kappa_{15}} (16 \binom{(4)}{\kappa_{16}} - 8 \binom{(4)}{\kappa_{16}} (2 \binom{(4)}{\kappa_3} - 3 \binom{(4)}{\kappa_7}) - \binom{(4)}{\kappa_3} (2 \binom{(4)}{\kappa_3} + 3 \binom{(4)}{\kappa_7})^2 + \binom{(4)}{\kappa_{16}} (-20 \binom{(4)}{\kappa_3}^2 - 24 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7} + 9 \binom{(4)}{\kappa_7}^2)) p^2 + \\ & \sqrt{3} \sqrt{((2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2 (4096 \binom{(4)}{\kappa_{15}}^6 p^4 + 4096 \binom{(4)}{\kappa_{15}}^5 (2 \binom{(4)}{\kappa_{16}} - \binom{(4)}{\kappa_3} + 2 \binom{(4)}{\kappa_7}) p^4 + \\ & \binom{(4)}{\kappa_{16}} (3 \binom{(4)}{\kappa_3} + 512 \binom{(4)}{\kappa_3} p^4) + 4 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} (15 \binom{(4)}{\kappa_3}^2 + 64 \binom{(4)}{\kappa_3} (6 \binom{(4)}{\kappa_3} + 5 \binom{(4)}{\kappa_7}) p^4) + \\ & 16 \binom{(4)}{\kappa_{15}} (147 \binom{(4)}{\kappa_3}^2 + 64 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_3} + 12 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_7} - 8 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7} + 6 \binom{(4)}{\kappa_7}^2) p^4) + \\ & 12 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3}^2 (33 \binom{(4)}{\kappa_3}^2 + 8 (20 \binom{(4)}{\kappa_3}^2 + 28 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7} + 13 \binom{(4)}{\kappa_7}^2) p^4) + \\ & 4 \binom{(4)}{\kappa_3} (48 \binom{(4)}{\kappa_3}^4 p^4 + 160 \binom{(4)}{\kappa_3}^3 \binom{(4)}{\kappa_7} p^4 + 72 \binom{(4)}{\kappa_3}^2 \binom{(4)}{\kappa_7}^2 p^4 + 27 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7}^3 p^4 + 24 \binom{(4)}{\kappa_3}^2 (8 \binom{(4)}{\kappa_3}^2 + 7 \binom{(4)}{\kappa_7}^2 p^4)) + \\ & 64 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3}^2 (16 \binom{(4)}{\kappa_3}^3 p^4 + 36 \binom{(4)}{\kappa_3}^2 \binom{(4)}{\kappa_7} p^4 + 9 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7}^2 p^4 + 3 \binom{(4)}{\kappa_3} (5 \binom{(4)}{\kappa_3}^2 + 8 \binom{(4)}{\kappa_7}^2 p^4)) + \\ & 32 \binom{(4)}{\kappa_{15}} (-84 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3}^2 + 128 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3}^3 p^4 + 64 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3}^4 p^4 + 288 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3}^2 \binom{(4)}{\kappa_7} p^4 + 160 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3}^2 \binom{(4)}{\kappa_7} p^4 + \\ & 72 \binom{(4)}{\kappa_7}^3 p^4 + 16 \binom{(4)}{\kappa_{16}} (16 \binom{(4)}{\kappa_3}^2 - 12 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7} + 15 \binom{(4)}{\kappa_7}^2) p^4 - 3 \binom{(4)}{\kappa_3} (77 \binom{(4)}{\kappa_3}^2 + 48 \binom{(4)}{\kappa_7}^2 p^4)) + \\ & 8 \binom{(4)}{\kappa_{15}} (256 \binom{(4)}{\kappa_{16}}^4 p^4 + 128 \binom{(4)}{\kappa_{16}}^3 (2 \binom{(4)}{\kappa_3} + 5 \binom{(4)}{\kappa_7}) p^4 + \binom{(4)}{\kappa_{16}} (738 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_3}^2 - 256 \binom{(4)}{\kappa_3}^3 p^4 + 768 \binom{(4)}{\kappa_3}^2 \binom{(4)}{\kappa_7} p^4 - 192 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7}^2 p^4 + 288 \binom{(4)}{\kappa_7}^3 p^4) + 3 \binom{(4)}{\kappa_{16}} (39 \binom{(4)}{\kappa_3}^2 + 16 (8 \binom{(4)}{\kappa_3}^2 + 4 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7} + 13 \binom{(4)}{\kappa_7}^2) p^4) + \\ & 2 (-80 \binom{(4)}{\kappa_3}^4 p^4 - 32 \binom{(4)}{\kappa_3}^3 \binom{(4)}{\kappa_7} p^4 - 72 \binom{(4)}{\kappa_3}^2 \binom{(4)}{\kappa_7}^2 p^4 + 27 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7}^3 p^4 + 3 \binom{(4)}{\kappa_3}^2 (177 \binom{(4)}{\kappa_3}^2 + 56 \binom{(4)}{\kappa_7}^2 p^4))) - \\ & 8 \binom{(4)}{\kappa_{15}} (256 \binom{(4)}{\kappa_{16}}^4 \binom{(4)}{\kappa_3} p^4 + 4 \binom{(4)}{\kappa_{16}}^3 (3 \binom{(4)}{\kappa_3}^2 + 160 \binom{(4)}{\kappa_3} (\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_7}) p^4) + \\ & 3 \binom{(4)}{\kappa_{16}}^2 \binom{(4)}{\kappa_3} (51 \binom{(4)}{\kappa_3}^2 + 16 (16 \binom{(4)}{\kappa_3}^2 + 22 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7} + 13 \binom{(4)}{\kappa_7}^2) p^4) + \\ & 2 \binom{(4)}{\kappa_3} (16 \binom{(4)}{\kappa_3}^4 p^4 + 112 \binom{(4)}{\kappa_3}^3 \binom{(4)}{\kappa_7} p^4 + 36 \binom{(4)}{\kappa_3}^2 \binom{(4)}{\kappa_7}^2 p^4 + 27 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7}^3 p^4 + 24 \binom{(4)}{\kappa_3}^2 (11 \binom{(4)}{\kappa_3}^2 + 6 \binom{(4)}{\kappa_7}^2 p^4)) + \\ & 2 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} (160 \binom{(4)}{\kappa_3}^3 p^4 + 480 \binom{(4)}{\kappa_3}^2 \binom{(4)}{\kappa_7} p^4 + 144 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_7}^2 p^4 + 3 \binom{(4)}{\kappa_3} (87 \binom{(4)}{\kappa_3}^2 + 88 \binom{(4)}{\kappa_7}^2 p^4)))))) / \\ & ((2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_3}) (-2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} + 2 \binom{(4)}{\kappa_3})^2 \\ & \binom{(2)}{\kappa_3}) \end{aligned}$

	1	$-\frac{3 \kappa_3^{(2)}}{3 \kappa_1 - 4 \kappa_{15} + 3 \kappa_2 + 2 \kappa_3} \kappa_3^{(4)}$	$-\frac{2}{3 \kappa_1 - 4 \kappa_{15} + 3 \kappa_2 + 2 \kappa_3} \kappa_3^{(4)}$
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	3	$\frac{2(-2\binom{(4)}{k_{15}}\binom{(4)}{k_{16}}\binom{(4)}{k_2}\binom{(2)}{k_3}}{2\binom{(4)}{k_{16}}^2-4\binom{(4)}{k_2}\binom{(4)}{k_3}+\binom{(4)}{k_7}+2\binom{(4)}{k_{16}}(-\binom{(4)}{k_2}+\binom{(4)}{k_3}+\binom{(4)}{k_7})+4\binom{(4)}{k_{15}}(\binom{(4)}{k_{16}}+\binom{(4)}{k_2}+\binom{(4)}{k_3}+\binom{(4)}{k_7})}$	$\frac{2(8\binom{(4)}{k_{15}}^2+6\binom{(4)}{k_{16}}^2+4\binom{(4)}{k_3}^2+4\binom{(4)}{k_3}\binom{(4)}{k_7}+3\binom{(4)}{k_7}^2+8\binom{(4)}{k_{15}}(\binom{(4)}{k_{16}}+\binom{(4)}{k_7})+8\binom{(4)}{k_{16}}(\binom{(4)}{k_3}+\binom{(4)}{k_7}))}{(2\binom{(4)}{k_{15}}\binom{(4)}{k_{16}}-2\binom{(4)}{k_3})(2\binom{(4)}{k_{16}}^2-4\binom{(4)}{k_2}\binom{(4)}{k_3}+\binom{(4)}{k_7}+2\binom{(4)}{k_{16}}(-\binom{(4)}{k_2}+\binom{(4)}{k_3}+\binom{(4)}{k_7})+4\binom{(4)}{k_{15}}(\binom{(4)}{k_{16}}+\binom{(4)}{k_2}+\binom{(4)}{k_3}+\binom{(4)}{k_7}))}$
Resolved unitarity condition(s):		(Demonstrably impossible)	